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Ever since the book has been freed, many volunteers have been busy translating it to numerous languages, such as Arabic, Brazilian Portuguese, German, Italian, Spanish, Japanese, Norwegian Bokmål, etc. Discover the full list of translations on the book's website: https://debian-handbook.info/get/#other

We would like to thank all the translators and translation reviewers. Your work is highly appreciated because it brings Debian into the hands of millions of persons who cannot read English.

## Personal Acknowledgments from Raphaël

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[3] http://dblatex.sourceforge.net

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## Keywords

**Objective**
**Means**
**Operation**
**Volunteer**



Chapter

# 1 The Debian Project

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*Before diving right into the technology, let us have a look at what the Debian Project is, its objectives, its means, and its operations.*

## 1.1. What Is Debian?

CULTURE

**Origin of the Debian name**

Look no further: Debian is not an acronym. This name is, in reality, a contraction of two first names: that of Ian Murdock, and his girlfriend at the time, Debra. Debra + Ian = Debian.

Debian is a GNU/Linux distribution. We will discuss what a distribution is in further detail in section 1.5, "The Role of Distributions" page 23, but for now, we will simply state that it is a complete operating system, including software and systems for installation and management, all based on the Linux kernel and free software (especially those from the GNU project).

When he created Debian, in 1993, under the leadership of the FSF, Ian Murdock had clear objectives, which he expressed in the *Debian Manifesto*. The free operating system that he sought would have to have two principal features. First, quality: Debian would be developed with the greatest care, to be worthy of the Linux kernel. It would also be a non-commercial distribution, sufficiently credible to compete with major commercial distributions. This double ambition would, in his eyes, only be achieved by opening the Debian development process just like that of Linux and the GNU project. Thus, peer review would continuously improve the product.

CULTURE

**GNU, the project of the FSF**

The GNU project is a range of free software developed, or sponsored, by the Free Software Foundation (FSF), originated by its iconic leader, Dr. Richard M. Stallman. GNU is a recursive acronym, standing for "GNU is Not Unix".

CULTURE

**Richard Stallman**

FSF's founder and author of the GPL license, Richard M. Stallman (often referred to by his initials, RMS) is a charismatic leader of the Free Software movement. Due to his uncompromising positions, he is not unanimously admired, but his non-technical contributions to Free Software (in particular, the legal and philosophical) are respected by everybody.

### 1.1.1. A Multi-Platform Operating System

COMMUNITY

**Ian Murdock's journey**

Ian Murdock, founder of the Debian project, was its first leader, from 1993 to 1996. After passing the baton to Bruce Perens, Ian took a less public role. He returned to working behind the scenes of the free software community, creating the Progeny company, with the intention of marketing a distribution derived from Debian. This venture was, sadly, a commercial failure, and development was abandoned. The company, after several years of scraping by, simply as a service provider, eventually filed for bankruptcy in April of 2007. Of the various projects initiated by Progeny, only *discover* still remains. It is an automatic hardware detection tool.

Ian Murdock died on 28 December 2015 in San Francisco after a series of worrying tweets where he reported having been assaulted by police. In July 2016 it was announced that his death had been ruled a suicide.

Debian, remaining true to its initial principles, has had so much success that, today, it has reached a tremendous size. Currently there are 10 hardware architectures officially supported and also other kernels like FreeBSD (although the FreeBSD-based ports are not part of the set of officially supported architectures). Furthermore, with more than 28,000 source packages, the available software can meet almost any need that one could have, whether at home or in the enterprise.

The sheer size of the distribution can be inconvenient: it is really unreasonable to distribute 16 DVD-ROMs to install a complete version on a standard PC... This is why Debian is increasingly considered as a "meta-distribution", from which one extracts more specific distributions intended for a particular public: Debian Science for scientific use, Debian Edu for education and pedagogical use in an academic environment, Debian Med for medical applications, Debian Jr. for young children, etc. A more complete list of the subprojects can be found in section 1.3.3.1, "Existing Debian Sub-Projects" page 18, dedicated to that purpose.

These partial views of Debian are organized in a well-defined framework, thus guaranteeing hassle-free compatibility between the various "sub-distributions". All of them follow the general planning for release of new versions. And since they build on the same foundations, they can be easily extended, completed, and personalized with applications available in the Debian repositories.

All the Debian tools operate in this direction: `debian-cd` has for a long time now allowed the creation of a set of CD-ROMs containing only a pre-selected set of packages; `debian-installer` is also a modular installer, easily adapted to special needs. `APT` will install packages from various origins, while guaranteeing the overall consistency of the system.

TOOL

**Creating a Debian CD-ROM**

`debian-cd` creates ISO images of installation media (CD, DVD, Blu-Ray, etc.) ready for use. Any matter regarding this software is discussed (in English) on the debian-cd@lists.debian.org mailing list. The team is led by Steve McIntyre who is handling official Debian ISO builds.

BACK TO BASICS

**To each computer, its architecture**

The term "architecture" indicates a type of computer (the most known include Mac or PC). Each architecture is differentiated primarily according to its processor, usually incompatible with other processors. These differences in hardware involve varying means of operation, thus requiring that software be compiled specifically for each architecture.

Most software available in Debian is written in portable programming languages: the same source code can be compiled for various architectures. In effect, an executable binary, always compiled for a specific architecture, will not usually function on any of the other architectures.

Remember that each program is created by writing source code; this source code is a text file composed of instructions in a given programming language. Before you can use the software, it is necessary to compile the source code, which means transforming the code into a binary (a series of machine instructions executable by the processor). Each programming language has a specific compiler to execute this operation (for example, `gcc` for the C programming language).

TOOL

**Installer**

`debian-installer` is the name of the Debian installation program. Its modular design allows it to be used in a broad range of installation scenarios. The development work is coordinated on the debian-boot@lists.debian.org mailing list under the direction of Cyril Brulebois.

### 1.1.2. The Quality of Free Software

Debian follows all of the principles of Free Software, and its new versions are not released until they are ready. Developers do not work upon a set schedule and don't have to rush to meet an arbitrary deadline. People frequently complain of the long time between Debian's stable releases, but this caution ensures that Debian's legendary reliability is met: long months of testing are indeed necessary for the full distribution to receive the "stable" label.

Debian will not compromise on quality: all known critical bugs on key packages are resolved in any new version, even if this requires the initially forecast release date to be pushed back. Optional packages whose critical bugs are not fixed, and thus do not meet the quality requirements, are simply dropped from the stable release.

### 1.1.3. The Legal Framework: A Non-Profit Organization

Legally speaking, Debian is a project managed by an American not-for-profit, volunteer association. The project has around a thousand *Debian developers*, but brings together a far greater number of contributors (translators, bug reporters, artists, casual developers, etc.).

To carry its mission to fruition, Debian has a large infrastructure, with many servers connected across the Internet, offered and hosted by many sponsors.

COMMUNITY

**Behind Debian, the SPI association, and local branches**

Debian doesn't own any server in its own name, since it is only a project within the *Software in the Public Interest* (SPI) association, which manages the hardware and financial aspects (donations, purchase of hardware, etc.). Although it was initially created specifically for the Debian project, this association now hosts other free software projects, especially the PostgreSQL database, Freedesktop.org (project for standardization of various parts of modern graphical desktop environments, such as GNOME and KDE Plasma), and the LibreOffice office suite.

➨ `https://www.spi-inc.org/`

In addition to SPI, various local associations collaborate closely with Debian in order to generate funds for Debian, without centralizing everything in the USA: they are known as "Trusted Organizations" in the Debian jargon. This setup avoids prohibitive international transfer costs, and fits well with the decentralized nature of the project.

Do not hesitate to join your local association and support the project!

➨ `https://wiki.debian.org/Teams/Auditor/Organizations`

➨ `https://france.debian.net/`

➨ `https://debian.ch/`

## 1.2. The Foundation Documents

A few years after its initial launch, Debian formalized the principles that it should follow as a free software project. This deliberately activist decision allows orderly and peaceful growth by ensuring that all members progress in the same direction. To become a Debian developer, any candidate must confirm and prove their support and adherence to the principles established in the project's Foundation Documents.

The development process is constantly debated, but these Foundation Documents are widely and consensually supported, thus rarely change. The Debian constitution also offers other guarantees for their stability: a three-quarter qualified majority is required to approve any amendment.

### 1.2.1. The Commitment towards Users

The project also has a "social contract". What place does such a text have in a project only intended for the development of an operating system? It is quite simple: Debian works for its users, and thus, by extension, for society. This contract summarizes the commitments that the project undertakes. Let us study them in greater detail:

1. Debian will remain 100% free.

   This is Rule No. 1. Debian is and will remain composed entirely and exclusively of free software. Additionally, all software development within the Debian project, itself, will be free.

   PERSPECTIVE
   **Beyond software**

   The first version of the Debian Social Contract said "Debian Will Remain 100% Free *Software*". The disappearance of this last word (with the ratification of Version 1.1 of the contract in April of 2004) indicates the will to achieve freedom, not only in software, but also in the documentation and any other element that Debian wishes to provide within its operating system.

   This change, which was only intended as editorial, has, in reality, had numerous consequences, especially with the removal of some problematic documentation. Furthermore, the increasing use of firmware in drivers poses problems: many are non-free, yet they are necessary for proper operation of the corresponding hardware.

2. We will give back to the free software community.

   Any improvement contributed by the Debian project to a work integrated in the distribution is sent back to the author of the work (called "upstream"). In general, Debian will cooperate with the community rather than work in isolation.

   COMMUNITY
   **Upstream author, or Debian developer?**

   The term "upstream author" means the author(s)/developer(s) of a work, those who write and develop it. On the other hand, a "Debian developer" uses an existing work to make it into a Debian package (the term "Debian maintainer" is better suited).

   In practice, there can be overlaps between both roles: the Debian maintainer may write a patch, which benefits all users of the work. In general,

Debian encourages those in charge of a package in Debian to get involved in "upstream" development as well (they become, then, contributors, without being confined to the role of simple users of a program).

3. We will not hide problems.

   Debian is not perfect, and, there will be new problems to fix every day. Debian will keep its entire bug report database open for public view at all times. Reports that people file on-line will promptly become visible to others.

4. Our priorities are our users and free software.

   This commitment is more difficult to define. Debian imposes, thus, a bias when a decision must be made, and will discard an easy solution for the developers that will jeopardize the user experience, opting for a more elegant solution, even if it is more difficult to implement. This means to take into account, as a priority, the interests of the users and free software.

5. Works that do not meet our free software standards.

   Debian accepts and understands that users may want to use some non-free programs. That is why the project allows usage of parts of its infrastructure to distribute Debian packages of non-free software that can safely be redistributed.

COMMUNITY

**For or against the non-free section?**

The commitment to maintain a structure to accommodate non-free software (i.e. the "non-free" section, see the sidebar "The `main`, `contrib` and `non-free` archives" page 109) is frequently a subject of debate within the Debian community.

Detractors argue that it turns people away from free software equivalents, and contradicts the principle of serving only the free software cause. Supporters flatly state that most of the non-free packages are "nearly free", and held back by only one or two annoying restrictions (the most common being the prohibition against commercial usage of the software). By distributing these works in the non-free branch, we indirectly explain to the author that their creation would be better known and more widely used if they could be included in the main section. They are, thus, politely invited to alter their license to serve this purpose.

After a first and unfruitful attempt in 2004, the complete removal of the non-free section is unlikely to return to the agenda, especially since it contains many useful documents that were moved simply because they did not meet the new requirements for the main section. This is especially the case for certain software documentation files issued by the GNU project (in particular, Emacs and Make).

The continued existence of the non-free section is a source of occasional friction with the Free Software Foundation, and is the main reason it refuses to officially recommend Debian as an operating system.

### 1.2.2. The Debian Free Software Guidelines

This reference document defines which software is "free enough" to be included in Debian. If a program's license is in accordance with these principles, it can be included in the main section;

on the contrary, and provided that free distribution is permitted, it may be found in the non-free section. The non-free section is not officially part of Debian; it is an added service provided to users.

More than a selection criteria for Debian, this text has become an authority on the subject of free software, and has served as the basis for the "Open Source Definition". Historically, it is therefore one of the first formal definitions of the concept of "free software".

The GNU General Public License, the BSD License, and the Artistic License are examples of traditional free licenses that follow the 9 points mentioned in this text. Below you will find the text as it is published on the Debian website.

➨ https://www.debian.org/social_contract#guidelines

1. **Free redistribution.** The license of a Debian component may not restrict any party from selling or giving away the software as a component of an aggregate software distribution containing programs from several different sources. The license may not require a royalty or other fee for such sale.

2. **Source code.** The program must include source code, and must allow distribution in source code as well as compiled form.

3. **Derived works.** The license must allow modifications and derived works, and must allow them to be distributed under the same terms as the license of the original software.

4. **Integrity of the author's source code.** The license may restrict source code from being distributed in modified form *only* if the license allows the distribution of "patch files" with the source code for the purpose of modifying the program at build time. The license must explicitly permit distribution of software built from modified source code. The license may require derived works to carry a different name or version number from the original software (*This is a compromise. The Debian group encourages all authors not to restrict any files, source or binary, from being modified*).

5. **No discrimination against persons or groups.** The license must not discriminate against any person or group of persons.

6. **No discrimination against fields of endeavor.** The license must not restrict anyone from making use of the program in a specific field of endeavor. For example, it may not restrict the program from being used in a business, or from being used for genetic research.

7. **Distribution of license.** The rights attached to the program must apply to all to whom the program is redistributed without the need for execution of an additional license by those parties.

8. **License must not be specific to Debian.** The rights attached to the program must not depend on the program being part of a Debian system. If the program is extracted from Debian and used or distributed without Debian but otherwise within the terms of the program's license, all parties to whom the program is redistributed should have the same rights as those that are granted in conjunction with the Debian system.

9. **License must not contaminate other software.** The license must not place restrictions on other software that is distributed along with the licensed software. For example, the license must not insist that all other programs distributed on the same medium must be free software.

BACK TO BASICS

**Copyleft**

Copyleft is a principle that consists in using copyrights to guarantee the freedom of a work and its derivatives, rather than restrict the rights of uses, as is the case with proprietary software. It is, also, a play of words on the term "copyright". Richard Stallman discovered the idea when a friend of his, fond of puns, wrote on an envelope addressed to him: "copyleft: all rights reversed". Copyleft imposes preservation of all initial liberties upon distribution of an original or modified version of a work (usually a program). It is, thus, not possible to distribute a program as proprietary software if it is derived from code from a copyleft released program.

The most well-known family of copyleft licenses is, of course, the GNU General Public License (GPL) and its derivatives, the GNU Lesser General Public License (LGPL), and the GNU Free Documentation License (GFDL). Sadly, the copyleft licenses are generally incompatible with each other. Consequently, it is best to use only one of them.

10. **Example licenses** The "GPL", "BSD", and "Artistic" licenses are examples of licenses that we consider "free".

BACK TO BASICS

**Free licenses**

The GNU GPL, the BSD license, and the Artistic License all comply with the Debian Free Software Guidelines, even though they are very different.

The GNU GPL, used and promoted by the FSF (Free Software Foundation), is the most common. Its main feature is that it also applies to any derived work that is redistributed: a program incorporating or using GPL code can only be distributed according to its terms. It prohibits, thus, any reuse in a proprietary application. This poses serious problems for the reuse of GPL code in free software incompatible with this license. As such, it is sometimes impossible to link a program published under another free software license with a library distributed under the GPL. On the other hand, this license is very solid in American law: FSF lawyers have participated in the drafting thereof, and have often forced violators to reach an amicable agreement with the FSF without going to court.

➨ `https://www.gnu.org/copyleft/gpl.html`

The BSD license is the least restrictive: everything is permitted, including use of modified BSD code in a proprietary application.

➨ `https://www.opensource.org/licenses/bsd-license.php`

Finally, the Artistic License reaches a compromise between these two others: integration of code in a proprietary application is permitted, but any modification must be published.

➨ `https://www.opensource.org/licenses/artistic-license-2.0.php`

The complete text of these licenses is available in `/usr/share/common-licenses/` on any Debian system (in case of BSD the newer 3-Clause License).

COMMUNITY

**Bruce Perens, a controversial leader**

Bruce Perens was the second leader of the Debian project, just after Ian Murdock. He was very controversial in his dynamic and authoritarian methods. He, nevertheless, remains an important contributor to Debian, to whom Debian is especially